

Srinidhi Shetty

+91 8310154673 | iamshettysrinidhi@gmail.com | [linkedin.com/in/Srinidhi-Shetty](https://www.linkedin.com/in/Srinidhi-Shetty) | [portfolio-srinidhishetty](https://portfolio-srinidhishetty.github.io)

EDUCATION

Dayananda Sagar College of Engineering

Bangalore, India

Bachelor of Engineering in Electronics and Communication Grade: 7.16/10

July 2023 – Sep 2027

Relevant Coursework: FPGA Prototyping, ASIC Design, VLSI Design, Digital System Design, Microprocessors & Microcontrollers, Signals & Systems, Embedded Systems, CMOS Technology, Semiconductor Physics.

Excellent PU College

Udupi, India

Science: (Physics, Chemistry, Mathematics, Computer Science) Grade: 95.3%

2021 – 2023

EXPERIENCE

Hardware R&D Engineer Intern

June 2026 – Present

MY Yuno AI

Bangalore, India

- Contributed to the research, design, and development of the hardware prototype for the AI companion device.
- Worked on hardware selection, prototyping, and system-level integration to support product development.
- Collaborated on hardware-software integration, ensuring seamless communication between embedded hardware and AI-driven software components.
- Worked closely with a multidisciplinary team to iterate on prototype design and translate product concepts into functional hardware.

Semiconductor Research Trainee

Dec. 2025 – Jan. 2026

Lam Research India – CeNSE, Indian Institute of Science (IISc)

Bangalore, India

- Gained hands-on exposure to semiconductor fabrication workflows using SEMulator3D, a process simulation platform widely used in semiconductor technology development.
- Developed an understanding of fabrication process flows, process integration, and advanced semiconductor manufacturing concepts.
- Strengthened foundational knowledge in semiconductor devices, fabrication technologies, and process engineering.

Hardware Prototyping Lead

May 2025 – Sep. 2025

Samsung Solve for Tomorrow 2025

Bangalore, India

- Led the hardware design and prototyping efforts for an interdisciplinary innovation project selected among the National Top 50 teams in Samsung Solve for Tomorrow 2025.
- Designed and developed the functional hardware prototype, translating conceptual ideas into a working proof of concept.
- Contributed to technical planning, system architecture discussions, and prototype validation throughout the competition.

PROJECTS

Verilog-Based AES-128 Hardware Encryption Engine | Verilog, ModelSim, Vivado

2024

- Designed and implemented a synthesizable AES-128 encryption engine in Verilog, incorporating the complete encryption pipeline including SubBytes, ShiftRows, MixColumns, and AddRoundKey transformations.
- Performed RTL simulation and functional verification using ModelSim.
- Synthesized the design for FPGA implementation using Vivado, analyzing timing performance and hardware testing.
- Strengthened understanding of digital hardware design and RTL verification methodologies.

Multi Clock Domain Controller IC | System Verilog, Quartus Prime, EDA Playground

2025

- Designed a Multi Clock Domain Controller (MCDC) to enable reliable data transfer between asynchronous clock domains.
- Implemented Clock Domain Crossing (CDC) techniques, including two-stage flip-flop synchronizers mitigate metastability.
- Developed robust handshaking mechanisms for safe inter-domain communication and data integrity.
- Verified functional correctness and timing behaviour through simulation, and running different testcases via the testbench codes to check and validate the functionality.

Smart Electric Vehicle Energy Network (SEVEN) | *Peer-to-Peer EV Energy Sharing Platform* 2024

- Proposed a peer-to-peer electric vehicle energy sharing platform enabling energy exchange between EVs to improve energy accessibility and reduce range anxiety.
- Designed a system architecture incorporating battery health monitoring, state-of-charge estimation, user authentication, and intelligent charging station discovery.
- Researched EV charging infrastructure, V2V/V2G technologies, and sustainable energy management strategies to support future mobility ecosystems.

Drishti – AI Smart Glasses for the Visually Impaired | *System Verilog, Quartus Prime, EDA Playground* 2024

- Designed a Multi Clock Domain Controller (MCDC) to enable reliable data transfer between asynchronous clock domains.
- Implemented Clock Domain Crossing (CDC) techniques, including two-stage flip-flop synchronizers mitigate metastability.
- Developed robust handshaking mechanisms for safe inter-domain communication and data integrity.
- Verified functional correctness and timing behaviour through simulation, and running different testcases via the testbench codes to check and validate the functionality.

TECHNICAL SKILLS

Hardware Description Languages: Verilog HDL, System Verilog.

Programming Languages: Embedded, C/C++, Python(Basics).

EDA & Simulation Tools: ModelSim, Vivado, Quartus Prime, Cadence Virtuoso, EDA Playground, Matlab.

Operating System & Tools: VS Code, Linux, PyCharm, tcl Scripting.

Domains: FPGA Prototyping, RTL Design & Verification, VLSI, Embedded Systems, ASIC Design, CMOS Technology, Physical Design, Hardware Prototyping, Product Design.

ACHIEVEMENTS & AWARDS

National Top 50 (Team Drishti) – Samsung Solve for Tomorrow 2025, hosted by Samsung India under the theme *Environment and Sustainability*.

1st Place (Gold Medal) – Nukkad Natak Competition, IIIT Kharagpur Spring Fest 2026, demonstrating cross-disciplinary excellence.

Special Domain Award – EthicsCare Hardware Hackathon, hosted by DSCE, for innovative hardware design under ethical computing constraints.

Runners-Up – Electronics 1.0 Hardware Hackathon, hosted by IEEE DSCE Chapter and VinyanaTech.

RESEARCH INTERESTS

FPGA Applications • Detector Electronics • Sensor Calibration • Digital Hardware Design • Semiconductor Physics • Solid State Physics

PATENTS

“Low Cost, Vehicle Independent, Driver Calibrated Brake Drag Detection and Alarm System”

Indian Patent Application – Filed (Under Examination)

Designed an embedded hardware system for real-time brake drag detection using sensor fusion and signal processing techniques for continuous brake drag monitoring and early fault detection.

CERTIFICATIONS

Semiconductor Skills Development Program through SEMulator3D

Lam Research India – Centre for Nano Science and Engineering (CeNSE), Indian Institute of Science (IISc)

CMOS Digital VLSI Design

NPTEL Online Certification, Indian Institute of Technology (IIT) Roorkee

LANGUAGES

English – Full Professional Proficiency

Hindi – Professional Working Proficiency

Kannada – Native or bilingual Proficiency